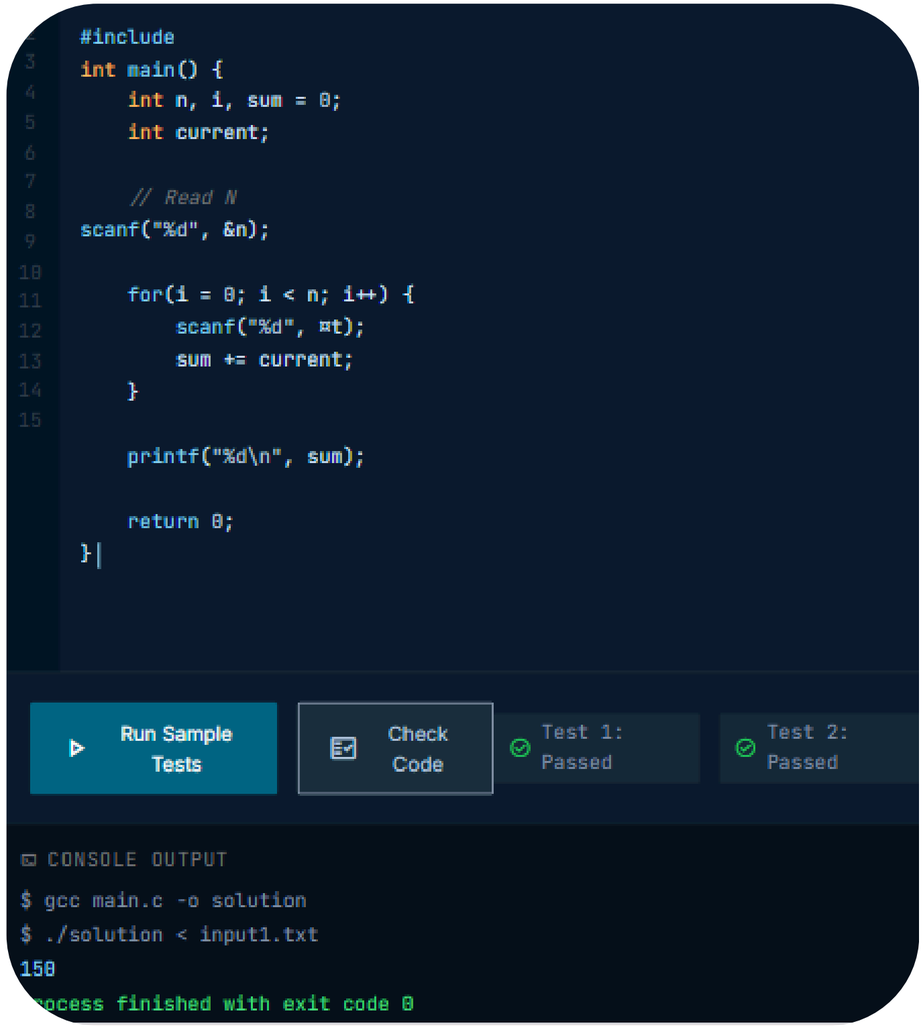
Cont/x

A Low-Level Keyboard-Hook Lockdown and On-Device Compiler-Based Assessment Engine for Secure Programming Exams



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The Problem

No Desktop Lockdown

Cloud tools only restrict browsers — Alt+Tab, Task Manager, and OS-level shortcuts remain open.

Server-Side Bottlenecks

Simultaneous submissions to remote grading servers cause delays during time-limited exams.

Unreliable Connectivity

Many Philippine schools face unstable internet, making cloud-only platforms risky for live exams.

No Unified Platform

No single tool combines system-level security AND automated code assessment together.

Proposed Solution

Cont/x

A Low-Level Keyboard-Hook Lockdown and On-Device Compiler-Based Assessment Engine for Secure Programming Exams

System-Wide Lockdown

Intercepts Alt+Tab, Windows key, Task Manager, copy-paste, screenshots at OS level

Local Code Compiler

Compiles and executes C programs locally via GCC/MinGW — no server needed

Teacher Dashboard

Live monitoring, violation alerts, and session control from any browser

Multi-Type Exam Builder

Multiple Choice, Fill-in-the-Blank, and Coding questions in one exam

Cloud Sync & Storage

Centralized records, submissions, and violation logs stored securely

Export Reports

CSV result sheets and incident reports generated instantly after the exam

Lockdown & Security

What's Intercepted & Blocked

- Alt+Tab and all app-switching shortcuts
- Windows key and Start menu
- Task Manager (Ctrl+Shift+Esc, Ctrl+Alt+Del)
- Copy-paste (Ctrl+C / Ctrl+V / Ctrl+X)
- Screenshot tools (PrintScreen, Win+Shift+S)
- Disallowed applications (process monitoring)
- System audio (muted on launch)

How It Works

1. When student starts their exam session → lockdown activates simultaneously on their machines
2. SetWindowsHookEx (WH_KEYBOARD_LL) intercepts OS-level keyboard input in Ring 3 userspace
3. Fullscreen always-on-top kiosk window captures focus
4. Active process monitoring flags disallowed apps instantly
5. Bypass attempts silently logged with timestamps on teacher dashboard
6. Lockdown released only when the student logs out of the Desktop Application

Assessment Engine

1

Student Writes Code

Monaco Editor (VS Code engine)

2

Local GCC Compiles

Process isolation, time limits

3

Checks Against Samples

Visible test cases before submit

4

Hidden Test Cases Run

Teacher-defined, never visible

5

Final Submission

Point deduction per check attempt

6

Score Returned

Partial credit + attempt deductions